

The Business Case for Product Recommendations

Turning a vast catalog into a personal storefront

A decision briefing for business stakeholders

Presented by Your Name • June 2026



A bigger catalog than anyone can browse — and rising expectations



Discovery is the bottleneck

Most of our catalog is never seen; shoppers can't buy what they never find.



Choice overload stalls purchases

Endless undifferentiated options delay decisions and depress conversion.



Personalization is now expected

71% of consumers expect personalized interactions; 76% are frustrated when it's missing.

What AI recommendations drive today

17%

of global online orders driven
by AI — recommendations,
promotions & service

+7%

higher average order value for
retailers using generative AI

Salesforce Shopping Index, 2024 (1.5B shoppers).
Recommendations are how leaders convert a huge
catalog into sales.

Published lift ranges, applied transparently to our numbers

What the research shows

- ✓ Personalization most often lifts revenue 5–15% (sector range up to ~25%)
- ✓ Can reduce customer-acquisition cost by up to 50%
- ✓ Improves marketing ROI by 10–30%
- ✓ Fast-growing firms earn ~40% more of their revenue from personalization

All figures: McKinsey (2021).

Illustrative on \$100M annual revenue

Applying the 5–15% revenue-lift range:

5% lift

\$5M

/year

10% lift

\$10M

/year

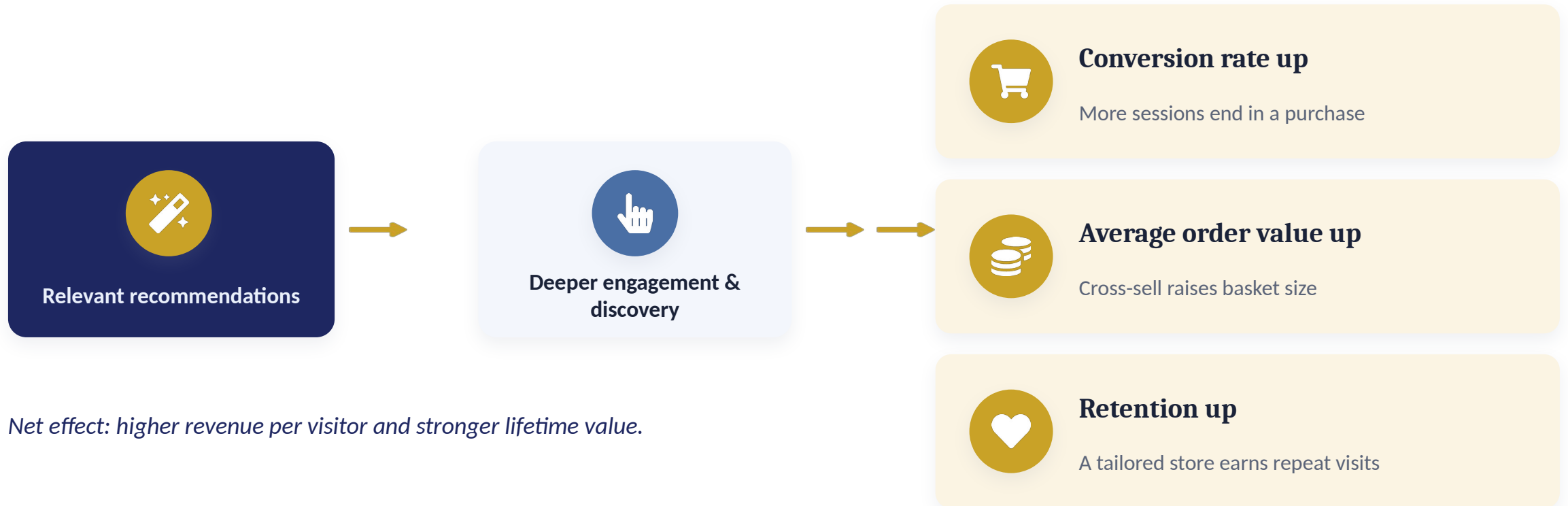
15% lift

\$15M

/year

Replace \$100M with our actual revenue. The percentages are sourced; the base is a placeholder for modeling, not a forecast.

How recommendations reach the numbers we report



Five surfaces, each with a business job



Personalized feed

Drives engagement

“For you” home and category feeds



Frequently bought together

Grows basket size

Cross-sell at cart & product page



Trending & popular

Converts undecided shoppers

Strong default with little data



Similar items

Extends the session

“More like this” keeps browsing



Recently viewed

Recovers lost intent

Easy path back to consideration

Learn from behavior, predict relevance, rank



The system learns from what shoppers browse and buy, predicts what each person is most likely to want next, and shows those items first — automatically, for every visitor.



What we know

Shopper history, product details, and the moment (device, time).



Smart matching

Scores how relevant each product is to this shopper.



Best picks shown

A short, ranked list on the page — refreshed continuously.

Market leaders treat recommendations as core infrastructure



Amazon

Pioneered item-to-item recommendations (“customers who bought this also bought”), launched 1998 and published in IEEE Internet Computing.

A foundational cross-sell engine, copied across the web.



Taobao / Alibaba

“Guess What You Like” home feed uses graph embeddings (EGES) to personalize a billion-item catalog; A/B-tested CTR gains.

Behavior-driven discovery at marketplace scale.



JD.com (Jingdong)

Deep-learning semantic retrieval and re-ranking serve hundreds of millions of users at ~\$100B+ annual GMV.

Relevance tied directly to sales at huge scale.

Adopting recommendations on our platform



Strengths

- We already own first-party behavioral data — the fuel for recommendations
- Scales personalization to every visitor automatically
- Mature, proven technology with vendor and open-source options
- Directly targets conversion, AOV, and retention



Weaknesses

- Cold start: thin signal for new users and items at launch
- Needs data quality, ML talent, and low-latency serving
- Results depend on traffic volume — thin data, weaker lift
- Ongoing maintenance and experimentation overhead



Opportunities

- 5–15% revenue lift typical (McKinsey, 2021)
- Differentiation: 71% of consumers expect personalization
- Cross-sell and surfacing of long-tail catalog
- Lower acquisition cost (up to ~50%) via better retention



Threats

- Privacy regulation and consent (e.g., GDPR, CCPA)
- Done poorly, personalization can cut revenue ~5% (McKinsey)
- Competitors with mature systems raise the bar
- Bias, filter bubbles, and reputational risk if unmanaged

What we'll actively control as we scale



Filter bubbles

Over-narrowing hurts discovery.

Mitigation: Inject diversity / variety controls.



Popularity bias

Best-sellers crowd out the long tail.

Mitigation: Debias training; promote fresh items.



Privacy & consent

Regulatory and trust exposure.

Mitigation: Consent-aware, data-minimized design.



Fairness

Uneven exposure across groups/sellers.

Mitigation: Audit exposure parity regularly.



Poor personalization

Done wrong, it can cut revenue ~5%.

Mitigation: Guardrails + A/B test every change.



Brand safety

Off-brand or non-compliant results.

Mitigation: Business rules and human review.

Prove value early, then scale what works



Phase 1 — Crawl

0–3 months

Popularity, trending, and rules-based
“bought together.”

Quick wins, low cost



Phase 2 — Walk

3–9 months

Add behavioral collaborative filtering and
similar-item recall.

Personalization begins



Phase 3 — Run

9–18 months

Personalized feeds, hybrid models, real-time
ranking.

Compounding returns

Each phase ships value on its own and gates the next on measured lift. The next slide shows a real platform that followed this exact path.

How Amazon actually walked this path

Amazon's own published history maps directly onto the crawl → walk → run plan.

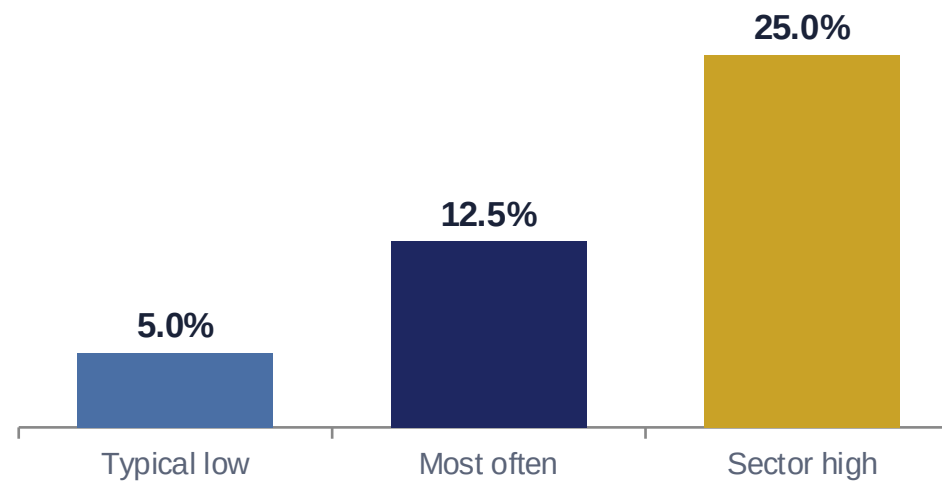


No claim ships without a controlled test

The method

- 1 Show recommendations to a random half of traffic (A/B test)
- 2 Compare against the holdout over a fixed window
- 3 Measure conversion, AOV, and revenue per session
- 4 Roll out only if lift is positive and statistically sound

What “good” looks like (published benchmark)



Personalization revenue-lift range — McKinsey (2021): most often 10–15%, sector range 5–25%.

THE ASK

What we'd like to move forward



Approve a Phase 1 pilot

A scoped, time-boxed test on one or two high-traffic surfaces.



Fund the team & tooling

A small cross-functional team plus experimentation tooling.



Agree success criteria

A clear lift target (benchmark: McKinsey 5–15%) that gates Phase 2.

KEY TAKEAWAYS

What to carry out of this session

- 1 Recommendations turn a vast catalog into a personal storefront
- 2 Published research: personalization typically lifts revenue 5–15% (McKinsey)
- 3 Market leaders (Amazon, Taobao, JD.com) treat them as core infrastructure
- 4 Phase the investment along Amazon's proven crawl → walk → run path
- 5 Prove every claim with A/B tests; manage privacy, bias, and fairness

Let's approve a focused pilot and measure the lift.

Thank you

Every figure in this deck is publicly sourced

Salesforce Shopping Index (2024)	Holiday-peak analysis of 1.5B shoppers. — AI (incl. personalized recommendations) drove 17% of global online orders; +7% AOV for retailers using generative AI. <i>salesforce.com/news/stories/ai-for-holiday-shopping-predictions</i>
McKinsey & Company (2021)	“The value of getting personalization right—or wrong—is multiplying.” — 5–15% revenue lift; CAC up to –50%; marketing ROI +10–30%; 71%/76% expectation; ~5% downside if done poorly. <i>mckinsey.com/capabilities/growth-marketing-and-sales/our-insights</i>
Gomez-Uribe & Hunt (2015)	“The Netflix Recommender System: Algorithms, Business Value, and Innovation,” ACM TMIS. — Recommendations estimated worth ~\$1B/year via reduced churn. <i>ACM Transactions on Management Information Systems</i>, 6(4)
Smith & Linden (2017); Linden et al. (2003)	“Two Decades of Recommender Systems at Amazon.com” and “Amazon.com Recommendations: Item-to-Item Collaborative Filtering,” IEEE Internet Computing. <i>ieeexplore.ieee.org/document/7927889</i>
Wang et al. (2018)	“Billion-scale Commodity Embedding for E-commerce Recommendation in Alibaba,” KDD 2018 — Taobao EGES graph embeddings; A/B-tested CTR gains. <i>arxiv.org/abs/1803.02349</i>
JD.com — Li et al. (2021); Zhao et al. (2017)	“From Semantic Retrieval to Pairwise Ranking”; JD scale (hundreds of millions of users, ~\$100B GMV). <i>arxiv.org/abs/2103.12982 · arxiv.org/abs/1709.00300</i>